Research Matrix Analysis

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Component	Thesis 1	Thesis 2	Thesis 3	Thesis 4	Thesis 5
Undergraduate/Master's/ Doctorate	BSCSSE	BSCSSE	Various	Various	
Title/Author/Year	GO CART: AI-Powered Shopping Cart With Self-Checkout System For Improved Efficiency And Personalized Grocery Shopping in the Philippines Domingo, Yves Rafael Itang, Ryan Caezar Pasion, Nikos Pastor, Kristel Angelo 2025	ENHANCING FITNESS: A MOBILE APPLICATION WITH BODY MOTION RECOGNITION AND REWARDING SYSTEM Arce, Inake Rafael L. Bustamante, Linus Aldrich Conception, Steven Miguel Sidhu, Ranjit Kent 2024	Artificial intelligence in dental radiology: a narrative review Ali, Muneeba Irfan, Memoona Ali, Tooba Wei, Calvin R. Akilimali, Aymar 2025	A Benchmark Dataset for Automatic Cephalometric Landmark Detection and CVM Stage Classification Khalid, Muhammad Anwaar Zulfiqar, Kanwal Bashir, Ulfat Shaheen, Areeba Iqbal, Rida Rizwan, Zarnab Rizwan, Ghina Fraz, Muhammad Moazam 2025	

Institution	Information and Communications Technology Academy (iAcademy)	Information and Communications Technology Academy (iAcademy)	Department of research, Medical Research Circle (MedReC)	Islamic International Dental College, Riphah International University	
Research Problem/Objectives	Supermarket queues, inawareness of promotions, outdated checkout systems	To aid in tracking and monitoring of the users' workout progress to efficiently help with the users' physical health	Assess the effects of artificial intelligence on dental radiology	Supplement Cephalometric dataset for future AI training endeavours	
Theoretical/Conceptual Framework	Electronic shopping cart that integrates computer vision and machine learning for object recognition and barcode scanning	Mobile fitness application designed to elevate workout experience through motion recognition, real-time pose detection, rewarding systems, progress tracking, and personalized user profiles.	Synthesize and outline main themes and trends of the use of AI in dental radiology based on existing literature	A dataset that features 29 most commonly used anatomical landmarks, with 15 skeletal, 8 dental, and 6 soft-tissue landmarks, annotated by a team of 6 skilled orthodontists in two phases, following extensive labelling and reviewing protocols.	
Research Design	RAD methodology	Agile methodology	Narrative review	Benchmark dataset creation	
Participants/Sample	Thirty participants	Eighteen to forty years old	N/A	1000 patients ranging from 12	

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	ranging from young adults to elderly			to 62 years old	
Data Collection Methods	Gathering and analyzing information on various AI frameworks different sensor options, and relevant software tools	Interview, Survey, and Secondary Data Analysis	Gathering relevant existing literature	The dataset consisted of lateral cephalometric radiographs retrieved from the orthodontic patient archives of Islamic International Dental College, Riphah International University, Islamabad, Pakistan	
Instrument Used	Visual Studio Code 1.91 Python 3.12 Supabase v1 React Native YOLOV8 CVAT Expo Next JS	Google ML Kit Java Firebase Firestore Android Studio Git	Radiology image modalities and datasets	X-rays used: ART Plus BLUEX, Veraviewepocs 2D J. Morita, Smart3D LargeV, ProMax 2D Planmeca, ProMax with ProTouch Planmeca, Hyperion X5 Myray, Rotograph EVO Villa	
Data Analysis Techniques	n/a	n/a	Reviewing of various techniques within the	Manual annotation of each x-ray with additional	

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			studies	validation by experts	
Key Findings	Project successfully developed a prototype of an AI-powered shopping cart that integrates computer vision, IoT, and self-checkout functionalities	Although the project experienced limitations, the project has successfully developed a mobile fitness application that engaged and motivated its users.	Artificial intelligence can improve clinical dental practice concerning dental radiography by improving diagnosis accuracy, thereby changing treatment planning and diagnostic procedures.	A dataset that boasts a diverse and extensive collection of 1000 cephalograms acquired from 7 different X-ray imaging devices with varying resolutions, making it the most comprehensive cephalometric dataset to date.	
Limitations/Gaps	The application has a relatively small dataset of eight products. It may limit the model's generalizability and accuracy in diverse real-world scenarios.	The application can only be used by users with a minimum android eight version of with at least 4GB of RAM and an 8MP front camera, The application will only be able to count the repitions of basic static exercioses, The application will not be able to detect a different form of an exercise.	The data pool of dental radiology is severely lacking in both quality and quantity. Additionally, inconsistent benchmarks, annotations, and assessments plague these studies which may hinder future studies.	AI algorithms may struggle to generalize on such a diverse set of patients and be prone to overfit. There is a high probability that the trained model may contain unnecessary bias, suggesting that there is a limit to clinical applications merely with this dataset.	

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Relevance to your Study	Object recognition with AI	Motion recogition using LLMs	Narrative review of AI in dental radiology	Documentation of dataset to be used for training	

Appendix

A. Research Articles

Ali, M., Irfan, M., Ali, T., Wei, C. R., & Akilimali, A. (2025). Artificial intelligence in dental radiology: a narrative review. *Annals of Medicine and Surgery*, *87*(4), 2212–2217. https://doi.org/10.1097/ms9.0000000000003127

Khalid, M. A., Zulfiqar, K., Bashir, U., Shaheen, A., Iqbal, R., Rizwan, Z., Rizwan, G., & Fraz, M. M. (2025). A benchmark dataset for automatic cephalometric landmark detection and CVM stage classification. *Scientific Data*, *12*(1), 1336. https://doi.org/10.1038/s41597-025-05542-3